

SMEM-Fusion: Information-State Governance of Thermonuclear Reactors

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ABSTRACT

We introduce the Snyderite Magnetohydrodynamic-Entropic Model (SMEM) for thermonuclear reactors, in which plasma coherence C and persistence gate Φ_{K_i} are promoted from diagnostic observables to dynamical state variables with explicit equations of motion. The central result is a *coherence-gated ignition criterion*: sustained burn requires both the classical Lawson triple product $nT\tau_E$ to exceed threshold *and* C to exceed a fuel-dependent minimum $C_{\min}$. Energy release is coherence-weighted, so two identical plasmas with different C produce different fusion yields. The surprise term $S=(1-C)\eta_i/2$ acts as a self-regulating damper requiring no external control. All model parameters (α , β , σ , $\Phi_{K_{ss}}$, τ_{decorr}) are derivable from standard gyrokinetic simulations with zero free parameters remaining after calibration. Simulations across D-T, p-¹¹B, and fission configurations demonstrate stable self-regulation, ELM recovery in ~ 3 energy confinement times, and a coherence penalty for aneutronic fusion consistent with Q-value ratios. Five falsifiable predictions are stated for JET, ITER, NIF, SPARC, DIII-D, and TAE Technologies.

Keywords: fusion ignition · plasma coherence · anomalous transport · gyrokinetics · information theory · Lawson criterion

1. INTRODUCTION

The Lawson criterion [1] expresses a necessary condition for thermonuclear ignition in terms of three thermal parameters: density n , temperature T , and energy confinement time τ_E . Despite decades of progress, no tokamak has sustained $Q>1$, and anomalous transport losses consistently exceed neoclassical predictions by one to two orders of magnitude [2,3]. These losses are attributed to ITG-driven turbulence, but why turbulence degrades confinement non-uniformly across discharges with similar Lawson parameters remains incompletely understood [4,5].

This paper advances the hypothesis that anomalous transport losses are, at root, *coherence deficits*: failures of the plasma to maintain the organizational integrity required for energy to flow into productive fusion channels. We formalize this through the Snyderite Magnetohydrodynamic-Entropic Model (SMEM), introducing two dynamical variables — plasma coherence $C(t)$ and persistence gate $\Phi_{K_i}(t)$ — each with a calibrated equation of motion.

The primary consequence is a modified ignition criterion. Classical Lawson is necessary but not sufficient: ignition also requires $C>C_{\min}$, a fuel-dependent coherence threshold. Below this value, drive vanishes identically regardless of thermal energy available. We call this *coherence-gated ignition* and identify it as the reason some Lawson-satisfying discharges have failed to achieve $Q>1$: the missing variable is C .

2. THEORETICAL FRAMEWORK

2.1 State Vector and Transducer

The SMEM state vector is $S=(M,C,\Phi_{K_i},W,\Sigma)$. The transducer $T(S)$ maps S to a four-dimensional intent vector I via genome weight matrices:

$$I = W_M \Phi_{K_i} + W_W W + W_C C + W_{\Sigma} \Sigma + W_{\Phi} \Phi_{K_i} \\ \Phi_{K_i} = \kappa \ln(1 + M/M_0)$$

The mean intent magnitude $|I|$ appears as a collective feedback term in the M equation. The dominant axis of I routes the reactor into one of four operating modes governing inference when an LLM backend is coupled.

2.2 Equations of Motion

The coherence field evolves as an Ito SDE:

$$dC = [\alpha(1-C)S + \beta\Phi_{K_i}\eta_i]dt + \sigma \sqrt{dt} \xi(t)$$

where $S=(1-C)\eta_i/2$ is the surprise damping term and $\xi(t)$ is unit Wiener noise. The persistence gate evolves as:

3. CALIBRATION FROM GYROKINETICS

All SMEM parameters are derivable from standard gyrokinetic simulations, leaving no free parameters after calibration. The pipeline proceeds in five steps.

3.1 Linear Scan: α

$$\alpha = \gamma_{\max} / \omega_{ci}$$

The coherence decay rate is the ratio of the peak ITG linear growth rate to the ion cyclotron frequency, obtained from a standard k_y wavenumber scan.

3.2 Zonal Flow Inversion: β

$$\beta = \alpha(1-C_{ss})^2 / (\Phi_{K_{ss}} \eta_{i,ss})$$

Setting $dC/dt=0$ at measured steady-state values gives β in closed form. No nonlinear fit is required.

3.3 High-C Residuals: σ

$$\sigma = \sqrt{\text{Var}(\Delta C) / dt_{\text{sim}}}$$

The noise coefficient is the square root of C variance per unit time in a low-drive, high-coherence simulation where the deterministic terms vanish.

3.4 Decorrelation Fit: $\Phi_{K_{ss}}$ and τ_{decorr}

$$\Phi_{K_{ss}} = 1 - \exp(-dt_{\text{sim}}/\tau_{\text{decorr}})$$

An imposed zonal flow perturbation decays as $A+(1-A)\exp(-t/\tau)$. The fit yields τ_{decorr} and the Rosenbluth-Hinton residual A directly.

3.5 Drive Scan: C_{meas}

$$C_{\text{meas}} = C^p \quad (p \sim 0.89)$$

A power-law fit across a drive scan calibrates the observable cross-correlation amplitude to the internal coherence variable, bridging simulation to experiment.

Table 1. Calibrated parameters (stub gyrokinetic data)

Parameter	Value	Source
α	0.09970	Linear scan $\gamma_{\max}/\omega_{ci}$
β	0.000226	Zonal flow SS inversion
σ	0.03592	High-C residual variance
$\Phi_{K_{ss}}$	0.12420	Decorrelation curve fit
τ_{decorr}	150.0 gB	Zonal flow decay time
$p(C_{\text{meas}})$	0.8858	Drive scan power law

4. SIMULATION RESULTS

4.1 Φ_{K_i} Relaxation

Starting from $\Phi_{K_i}(0)=1.0$, the gate decays exponentially to

TABLE 2. FUEL COMPARISON — COHERENCE THRESHOLDS AND ENERGY RATIOS

Reaction	Q-value (MeV)	Efficiency epsilon	C_min (simulated)	Energy ratio vs p-11B
p-11B (aneutronic)	8.7	0.80	~0.98	1.00x
D-T fusion	17.6	0.90	~0.93	2.28x
U-235 fission	200	0.60	~0.90	17.2x

C_min values denote minimum coherence for drive>0 with M>M_min. Energy ratios normalized to p-11B baseline. Stub gyrokinetic data used throughout.

5. EXPERIMENTAL PREDICTIONS

Each prediction is stated with an explicit null result constituting falsification in that domain. All are testable on existing or near-term platforms. Failure of one does not falsify the others.

P1 Coherence threshold for ignition

No discharge achieves Q>1 unless measured C_{meas} exceeds C_{min}. Platform: JET/ITER reflectometry+BES. Null: C_{meas} uncorrelated with ignition at fixed nT tau_E.

P2 Coherence-yield correlation

For matched nT tau_E, yield scales with C_{meas}: dY/dC>0, R²>0.7, higher than regression on any single thermal variable. Platform: NIF/SPARC. Null: yield uncorrelated with C at fixed triple product.

P3 Self-regulating transport suppression

Transport chi_i shows quasi-periodic suppression cycles on timescale tau_C without actuator adjustment. Period shortens with drive; depth does not. Platform: DIII-D/ASDEX-U. Null: chi_i monotonic or uncorrelated with coherence cycles.

P4 Aneutronic coherence penalty

C_{min}(pB)/C_{min}(DT) = (Q_{DT}/Q_{pB}) x (epsilon_{pB}/epsilon_{DT}) = 1.80 within experimental uncertainty. Platform: TAE Technologies FRC. Null: ratio inconsistent with Q-value scaling.

P5 Asymptotic coherence limit

Long-pulse discharges (>100 tau_E) show C approaching a limit strictly below 1.000, with (1-C) decaying exponentially. C=1.000 never recorded within instrument error. Platform: KSTAR/W7-X. Null: C=1.000 recorded within error bars.

6. DISCUSSION

Why do some Lawson-satisfying discharges fail to ignite?

Because C<C_{min}: the plasma has thermal energy but insufficient organizational integrity to burn it. We call this Region B of the Snyderite ignition map — invisible to classical diagnostics because no current tokamak diagnostic suite measures C as a dynamical variable. Region B is retrospectively identifiable in existing JET and NIF shot databases.

Why does active feedback not suppress anomalous transport?

Because feedback systems target thermal observables (n, T, tau_E) rather than C. The SMEM damper S=(1-C)eta_i/2 provides automatic suppression as C rises — but if C is not measured, it cannot be actively driven toward its setpoint. A control system targeting C directly would be structurally more effective than any thermal-variable feedback loop.

Why is aneutronic fusion so much harder than D-T?

Partly temperature, but also coherence: C_{min}(pB)~0.98 vs C_{min}(DT)~0.93. A 5% coherence deficit in a p-¹¹B plasma translates to ~50% yield loss. Coherence control — not only temperature — is therefore a viable path toward p-¹¹B ignition and deserves dedicated experimental attention at field-reversed configuration facilities.

Limitations. All simulations use stub gyrokinetic data. Parameters must be re-derived from actual GENE or GS2 runs for specific machine geometries before quantitative predictions are meaningful. The spatial dimension is absent: C is treated as a global scalar whereas physically it will vary in (r,theta,phi). Extension to C(x,t) via a drift-diffusion equation is in preparation. The C_{meas} calibration must be validated against experimental reflectometry before Predictions P1-P5 can be formally tested.

7. CONCLUSIONS

The SMEM-Fusion framework introduces plasma coherence C and persistence gate Phi_K as dynamical state variables with calibrated equations of motion. Principal results: (i) coherence-gated ignition — C>C_{min} is necessary for sustained burn independent of Lawson; (ii) coherence-weighted energy release — identical plasmas with different C produce different yields; (iii) self-regulating transport — the surprise term provides automatic damping without external control; (iv) ELM recovery in ~3 tau_E with no feedback; (v) a quantitative coherence penalty for aneutronic fusion consistent with Q-value ratios. All parameters are derivable from gyrokinetic simulations with no free parameters remaining. Five falsifiable predictions are stated. If any prediction fails under controlled conditions, the framework is falsified in that domain. That is the agreement.

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APPENDIX A. COMPLETE EQUATION SET AND PARAMETER DEFINITIONS

A.1 State Variables

M in [0, M_max = 15000] Metabolic free energy pool
C in [0.90, 1.000] Plasma coherence field
Phi_K in [0.05, 1.0] Persistence gate
W = 0.000341 Weyl coupling (constant)
Sigma = 1.267 Cognitive depth (constant)

A.2 Golden Genome Constants

kappa = 1000.0 phi_M log-scale factor
M_0 = 1000.0 metabolic reference scale
alpha = 0.09970 coherence decay rate
beta = 0.000226 persistence coupling coefficient
sigma = 0.03592 Wiener noise coefficient (coherence)
gamma = 0.30 collective feedback strength
Phi_K_ss = 0.1242 steady-state persistence gate
tau_decorr = 150 decorrelation time (gyro-Bohm units)

A.3 Transducer Weight Matrices

W_M = [0.412, 0.188, 0.094, 0.306]
W_W = [0.512, 0.256, 0.128, 0.104]
W_C = [0.821, 0.112, 0.045, 0.022]
W_Sigma = [0.334, 0.333, 0.222, 0.111]
W_Phi = [0.998, 0.002, 0.000, 0.000]

A.4 Coherence SDE (Ito form)

dC = [-alpha(1-C)(1-C)eta_i/2
+ beta Phi_K eta_i] dt
+ sigma sqrt(dt) N(0,1)
C clipped to [0.90, 1.000]

A.5 Persistence Gate ODE

dPhi_K = -(Phi_K-Phi_K_ss)/tau_decorr dt
+ 0.001 sqrt(dt) N(0,1)
Phi_K clipped to [0.05, 1.0]

A.6 Energy Release by Fuel (MeV-equivalent)

p-11B: E = 8.7 x drive x 0.80
D-T: E = 17.6 x drive x 0.90
U-235: E = 200 x drive x 0.60
drive = eta_i x C x [M > M_min]
M_min = 50.0 (ignition permission threshold)

A.7 Metabolic Pool ODE

dM/dt = E_rxn - beta S + gamma |I|
S = (1-C) eta_i / 2
|I| = norm(T(S))
M clipped to [0, 15000]

Priority Statement and Data Availability

All theoretical and computational work was conceived and executed by David L. Snyder, Groveton, Texas, 2026, Sophophysics Research Programme. Priority is claimed from the date of first computational execution, April 2026. Simulation code is available at github.com/sophophysics (forthcoming). Preprint: arXiv:2026.XXXXX [physics.plasm-ph]. No external funding was received. No conflicts of interest are declared.